Unit 2 Assignment 4

Group Assignment (4–5 Students)

Objective

In this assignment, your team will perform a complete statistical analysis on a real-world dataset. You will explore the data, classify variables, analyze distributions, compare sampling methods, compute descriptive statistics, investigate relationships between variables, and interpret your findings before building a Machine Learning model.

Dataset

Use the **Titanic Dataset**.

Task 1 – Dataset Exploration

Load the dataset using Pandas.

Answer the following:

1. How many rows and columns does the dataset contain?
2. List all column names.
3. Display the data type of each variable.
4. Identify any missing values.
5. Briefly describe the dataset in 3–5 sentences.

Task 2 – Variable Classification

For each variable, identify:

- Numerical or Categorical

Then classify it as one of the following:

- Nominal
- Ordinal
- Discrete
- Continuous

Present your answer in a well-organized table.

Task 3 – Features and Target

Assume you want to predict whether a passenger survived.

Answer the following:

1. What is the target variable?
2. Which variables are considered input features?
3. Why is the target variable called the dependent variable?
4. Why are the remaining variables considered independent variables?

Task 4 – Population and Sampling

Assume the entire Titanic dataset represents the **population**.

Create the following samples:

- Random Sample (30%)
- Stratified Sample (30%) based on the **Survived** variable

Then answer:

1. What is the population?
2. What is the sample?
3. Compare the sizes of the population and the two samples.
4. Which sampling method better preserves the original class distribution?

5. Explain why representative sampling is important in Machine Learning.

Task 5 – Descriptive Statistics

Select **three numerical variables**.

Calculate:

- Mean
- Median
- Mode
- Minimum
- Maximum
- Range
- Variance
- Standard Deviation

Then answer:

1. Which variable has the highest average?
2. Which variable has the greatest variability?
3. Are the mean and median close?
4. What does this indicate about the data distribution?

Task 6 – Distribution and Outlier Analysis

Choose **two numerical variables**.

Create:

- Histogram
- Boxplot

Then answer:

1. Is the distribution approximately symmetric or skewed?
2. Are there any outliers?
3. Which statistic is more affected by outliers: Mean or Median?
4. Explain your answer using the plots.

Task 7 – Correlation and Covariance Analysis

Using only the numerical variables:

1. Compute the Correlation Matrix.
2. Visualize it using a Heatmap.
3. Compute the Covariance Matrix.

Then answer:

1. Which variables have the strongest positive correlation?
2. Which variables have little or no relationship?
3. What is the difference between Correlation and Covariance?
4. Which one is easier to interpret? Why?

Task 8 – Statistical Insights

Write **five observations** about the dataset based on your analysis.

Examples:

- Which variable has the largest spread?
- Which variable contains outliers?
- Which variables appear to be related?
- Which variable has the highest average?
- Which variable may require preprocessing?

Task 9 – Statistics for Machine Learning

Imagine you are preparing this dataset for a Machine Learning model.

Answer the following:

1. Which feature do you think will be most useful for predicting survival?
2. Which feature would you consider removing? Why?
3. Based on the Correlation and Covariance analysis, which variables do you expect to contribute the most if PCA were applied?
4. What preprocessing steps would you perform before training the model?

Support your answers using evidence from your statistical analysis.

Deliverables

Each group must submit:

- Jupyter Notebook (.ipynb)
- PDF Report